R18

Code No: 152AA

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year II Semester Examinations, July/August - 2021
MATHEMATICS-II
(Common to CE, EEE, ME, ECE, CSE, EIE, IT, MCT, MMT, AE, MIE, PTM, ITE)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Solve $x\frac{dy}{dx} + y = x^3y^6$.

b) Solve $(y - px)(p - 1) = p$; where $p = \frac{dy}{dx}$. [7+8]

2. Water at Temperature 100^0C cools in 10 minutes to 80^0C in a room temperature 25^0C. Find the temperature of water after 20 minutes. [15]

3. Solve $(D^2 - 4D + 4)y = x^2 \sin x + e^{2x} + 3$. [15]

4.a) Evaluate $\int_0^1 \int_1^{2-x} xydxdy$.

b) Find by double integration, the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. [7+8]

5.a) Evaluate $\iint_R ydxdy$, where R is the region bounded by the parabolas $y^2 = 4x$ and $x^2 = 4y$.

b) Evaluate $\int_0^{\pi} \int_0^{a \sin \theta} rdrd\theta$. [7+8]

6.a) Find the work done by the force $\vec{F} = 3x^2i + (2xz - y)j + zk$ along the straight line joining the points (0,0,1) and (2,1,3).

b) Find $curl\overline{f}$ where $\overline{f} = grad(x^3 + y^3 + z^3 - 3xyz)$. [8+7]

7.a) Find the directional derivative of the function $f = x^2 - y^2 + 2z^2$ at the point $P(1,2,3)$ in the direction of the line PQ where Q is the point $(5,0,4)$.

b) Prove that $div\left(\frac{\vec{r}}{r^3}\right) = 0$, where $\vec{r} = xi + yj + zk$. [8+7]

8. State and verify Gauss divergence theorem for $\overline{f} = (x^3 - yz)i - 2x^2yj + zk$ taken over the surface of the cube bounded by the planes $x = y = z = a$ and coordinate planes. [15]

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